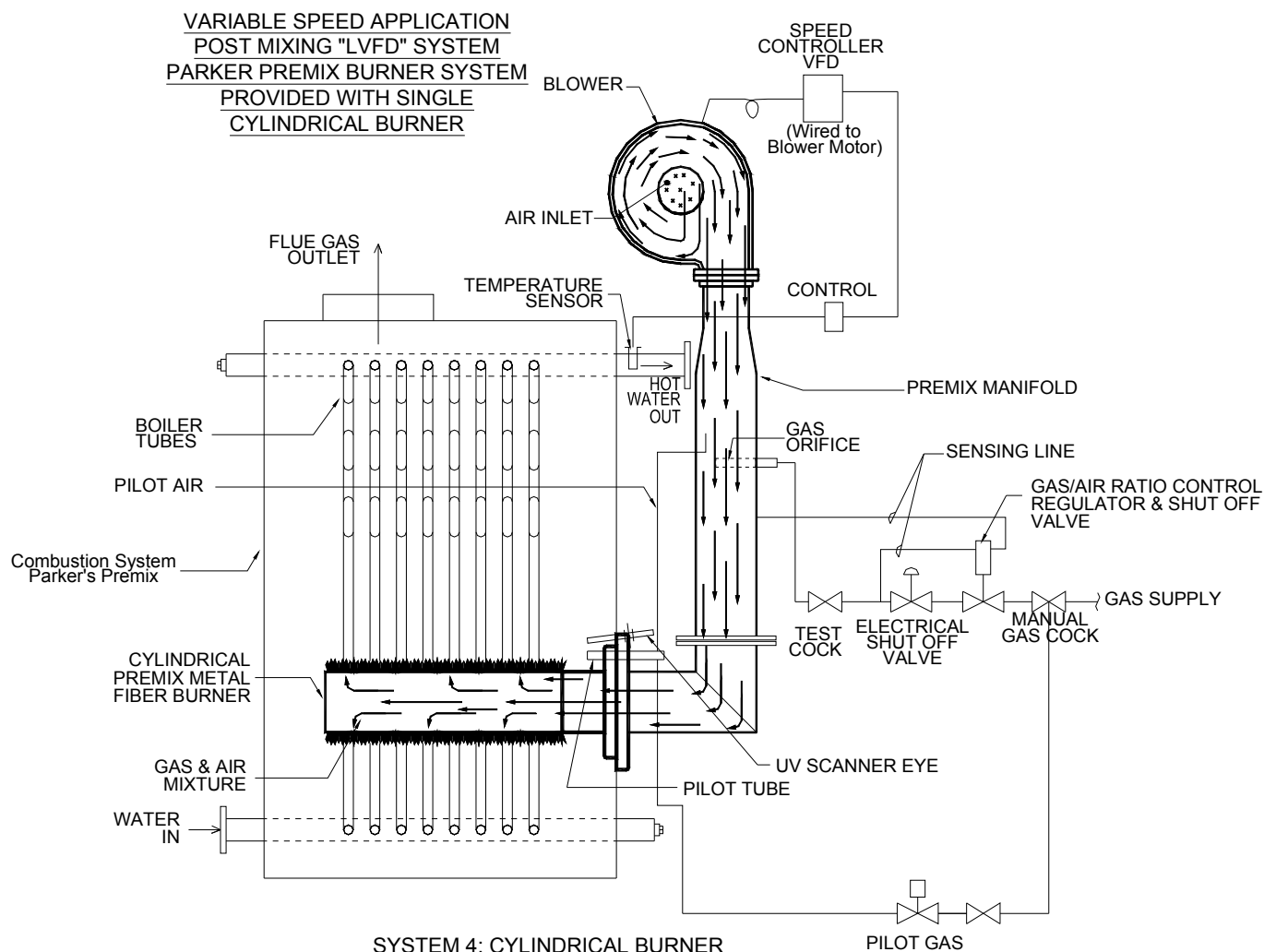


SECTION 3: BURNER SYSTEM OVERVIEW

3.1 – BURNER TYPES

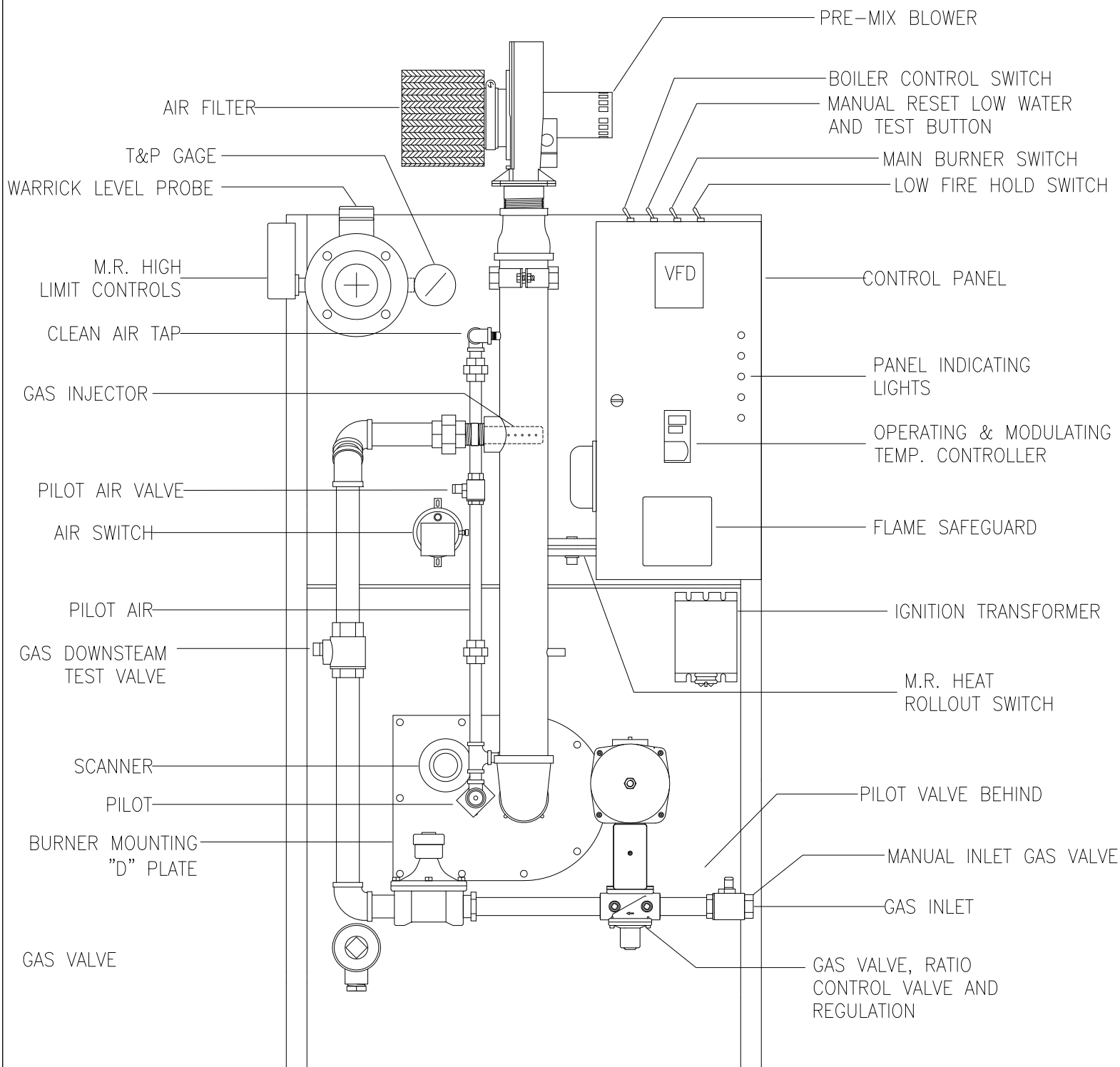
The boilers will be fitted with either a cylindrical Metal Fiber Burner for Low NOx Emissions on natural gas or LPG or a Powerflame burner for burning natural or LPG, gas, oil, or gas & oil.

3.2 – CYLINDRICAL METAL FIBER BURNER



PREMIX3-ww.DWG

WATERWALL CONTROL LAYOUT



FRONT VIEW

USED ON WATERWALL HOT WATER BOILER PART NAME WATERWALL CONTROL LAYOUT

FOR WATERWALL HOT WATER BOILERS

DR. FDL	DATE 12/7/11
CH.	
APPROVED	
SUPERSEDES NO.	

PARKER BOILER CO.

5930 BANDINI BLVD.

LOS ANGELES, CA 90040-2999

SCALE NTS
DIMENSIONS INCHES
FILE DESIGNATION PROPOSAL\204\
DWG. NO. WW Control Layout-G1008RL

PARKER-LITE SEQUENCE INDICATOR SYSTEM

For All Boilers



Standard Parker-Lite Sequence Indicator Systems

The PARKER-LITE SEQUENCE INDICATOR CONTROL SYSTEM features a multi-light indicator system whereby the operator can tell at a glance the exact operating stage of the boiler and immediately assess or isolate the cause of a failure. The panel and control system are expertly designed to offer a truly superior control panel and sequence indicator system operation. The Parker-Lite Sequence Indicator System is furnished standard on steam boilers 30 HP or larger and all boilers over 2.5 MM BTU. It is also furnished standard with FM trim and can be ordered on other models for a small additional cost.



COMPLETE SEQUENCE INDICATOR SYSTEM

The Parker-Lite System indicates actual operating conditions in the boiler. During normal operation, all lights will be on. Should any unsafe conditions occur, the main burner is immediately shut off and the appropriate light turns off indicating the problem area.

The light will automatically turn back on when a safe operating condition is re-established. Some newer flame safeguard systems provide additional diagnostic indicator lights so that eight or more conditions can be monitored when combined with the Parker-Lite.

INDICATOR LIGHTS

Several variations are available but the standard system is equipped with lights indicating these conditions:

- 1) Control Power On
- 2) Level Safe
- 3) Limit Safe
- 4) Pilot On
- 5) Burner On

or

- 1) Control Power On
 - 2) Level Safe
 - 3) Limit Safe
 - 4) Call for Heat
- The "Burner On" or "Call for Heat" and the "Pilot On" light, on interrupted and intermittent pilot systems, will cycle during normal operations. Other lights are available as an option.

ECONOMY

Service calls and down-time are minimized since the system itself saves time by immediately isolating the trouble condition that has shut off the boiler. The cause of a failure can often be isolated easily over the phone and the problem solved without a service call.

SAFETY

The boiler safety controls shut off the main burner with any failure and the indicator light turns off so the operator immediately knows where the failure has occurred. Because all lights are lit during normal operations burned out lights can be quickly spotted and replaced.

QUALITY CONSTRUCTION

The Parker-Lite Indicator System is wired with an electrical color code with each light having a different colored lead wire. Indicator lights are mounted on the control panel door to assure visibility from three directions.



"Never a Compromise for Quality or Safety"

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BROCHURE CP 0E7

3.4 – SEQUENCE OF OPERATION

GB6 OR GC6 CONTROL SYSTEM WITH M SERIES OR RM7800 SERIES PARKER-LITE SEQUENCE INDICATOR & INTERRUPTED PILOT

I. CONTROL POWER ON:

With the electrical switches turned "On", the indicator light "Control Power on" should be on. This indicates electrical power to the boiler controls and that the boiler is ready for operation.

If this does not come on, check switches, fuses or circuit breakers, improper electrical supply, broken, loose or dirty electrical connections. If a door interlock is furnished, be certain the boiler control panel door is tightly shut so that door interlock is energized.

II. WATER SAFE:

With the control power on a circuit to the water control is energized. If there is a safe water level in the boiler, the "Water Level Safe" indicator light will turn on. NOTE: If a manual reset type low water cutoff is used, it may first be necessary to reset the control before the light will turn on.

III. LIMIT SAFE:

With power on, and water level safe, the circuit to the limit controls is energized. There are a number of limit controls that must be safe (closed) to allow boiler operation they are:

- 1) High Temperature Limit (Manual Reset Type) or (Auto Reset)
- 2) Heat Rollout Switch (Manual Reset Type)
- 3) Flue Gas Spillage Switch if provided

If all limits are satisfied, Safe & Reset the "Limit Safe" indicator light will turn on. Note that other limits special ordered may also be present.

These three lights "Control Power", "Level Safe", and "Limit Safe" must be on before boiler will fire up.

IV. PILOT ON AND BURNER ON:

With power on, water level safe, limit safe, and the boiler controls and main burner switch on, a circuit is energized to the operating control. If the operating control & any BMS contacts are calling for heat, the circuit to the flame safeguard is completed the light off sequence takes place: Note that some boilers may be equipped with flow switches which will not allow boiler to fire unless flow is proven. Refer to Burner Control System Section 3.6

3.5 – OPERATING INSTRUCTIONS

PARKER GAS FIRED STEAM OR HOT WATER BOILERS WITH GB6 OR GC6 CONTROL SYSTEMS

I. STEPS IN STARTING THE BOILER:

1. CLOSE MAIN AND PILOT GAS COCKS AND WAIT 5 MINUTES.

2. TO LIGHT: (Do not attempt to light pilot with main downstream gas cock open).

- A. Turn on electricity. Turn off main burner switch. Open upstream gas cock, but leave main downstream gas cock closed. Manual reset controls, if furnished, must be reset.
- B. With control power on, water level safe, and limit safe, open small pilot gas cock and turn on main burner switch. The flame safeguard is energized and the burner startup sequence begins. See Sequence of Operation Sheet for complete details.

- C. The pilot will automatically ignite and the "Pilot On" light will turn on. With pilot safely on, immediately open the main downstream gas cock near the burner. If this cock is not opened, the system will turn off on safety lockout.
 - D. The "Burner On" light will turn on and the boiler is ready for normal operation. During normal operation the flame safeguard will supervise the main burner flame. The pilot and "Pilot On" light will be off.
3. **IF BURNER LOCKS OUT, RESET FLAME SAFEGUARD AND REPEAT ABOVE INSTRUCTIONS.**
For further details, see instructions below and Flame Safeguard Bulletin.
4. **DAILY ROUTINE CHECK ALL SAFETY CONTROLS.**

CAUTION: DO NOT MANUALLY HOLD IN FLAME SAFEGUARD; SERIOUS HAZARD RESULTS

WARNING: ONLY QUALIFIED SERVICE PERSONNEL SHOULD ATTEMPT TO START UP, ADJUST OR REPAIR THIS UNIT. NEVER BYPASS ANY SAFETY CONTROL.

II. SHUTDOWN INSTRUCTIONS:

1. Turn off the main burner switch.
2. Check flame to see that main and pilot burners are off. Check lights and for flame in back sight glass.
3. Manually close the main downstream gas cock nearest the burner.
4. Manually close the pilot gas cock.
5. Turn off electricity to boiler.

III. BURNER FAILURE INSTRUCTIONS:

If boiler ceases to fire or refuses to start:

1. Manually close main and pilot gas cocks, reset button on flame safeguard and follow above instructions.
2. If a pilot failure, low water condition or high limit condition has caused the boiler to shut down, do not place the boiler back in operation until the cause of the shutdown has been determined and corrected. If a flame failure occurs, always close main and pilot gas cocks and wait at least 5 minutes to assure an adequate purge before attempting to restart boiler.
3. Check water level, safety and limit controls.
4. Check electric supply and fuse box or circuit breakers.
5. Check gas supply.
6. The proper positioning of the scanner eye is very critical as it must be in a position to adequately detect both the pilot and main burner flame.
7. Review Sequence of Operation, Burner Control System and Flame Safeguard Instruction Sheets.
8. If the boiler still fails, call in competent service personnel.
9. Do not tamper with controls or adjustment except under supervision of qualified personnel.

NOTE: The main downstream gas cock is the one on the burner manifold after the electric gas valves nearest the burner.

3.6 - BURNER CONTROL SYSTEM

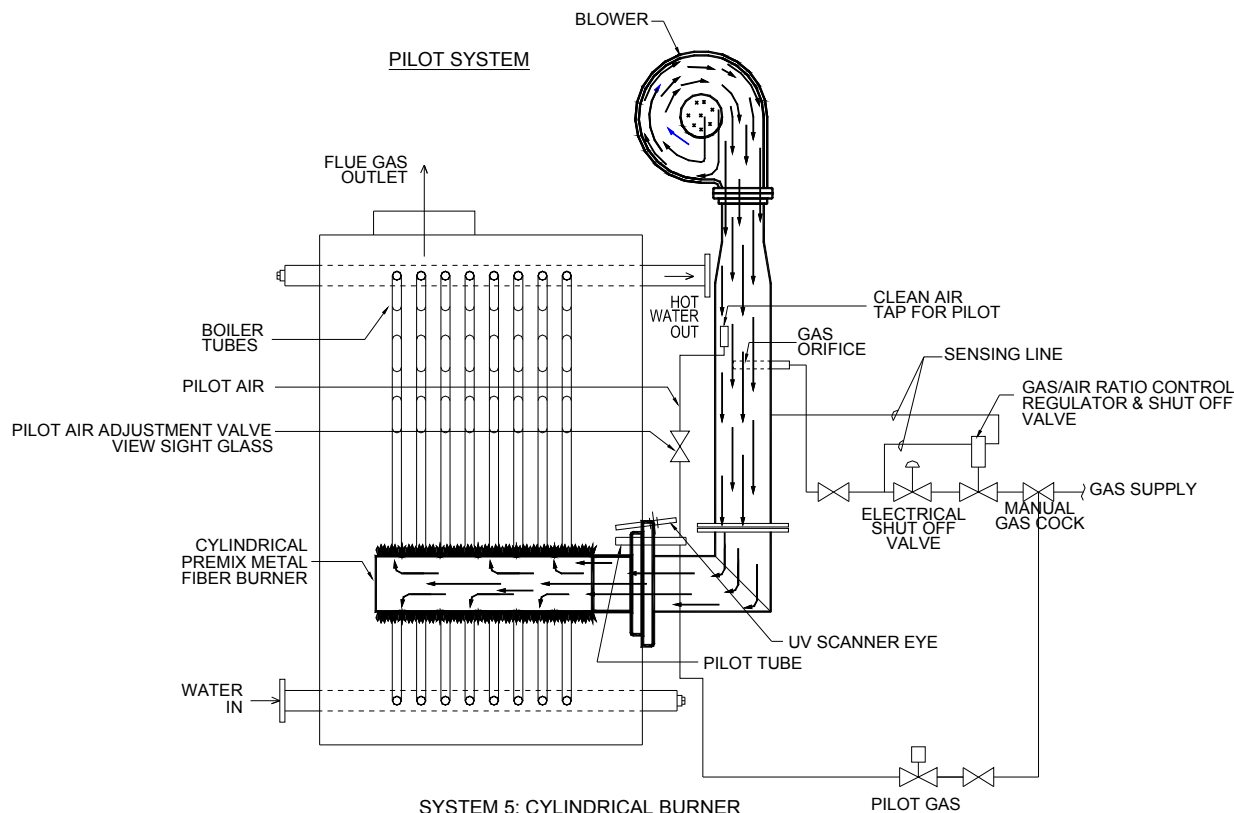
GC6 LN (LOW NO_x) PARKER GAS HOT WATER BOILERS FIREYE M SERIES OR HONEYWELL RM7800 SERIES FLAME SAFEGUARD WITH INTERRUPTED PILOT

1. FLAME SAFETY SYSTEM:

The Flame Safety System consists of a Flame Safeguard (Fireye M Series with programmer or a Honeywell RM7800 Series) with a UV Scanner to provide instant automatic boiler shut-off on pilot or main flame failure. The electronic combustion safeguard control is designed for flame supervision. The UV Scanner is designed for instantaneous flame detection. The UV Scanner must detect the pilot flame within 10 seconds of initiation of pilot ignition sequence or the Flame Safeguard will "Lock-Out" the burner and indicate an alarm condition for the burner. This can be cleared by pressing the reset button on the Flame Safeguard. If the pilot properly lights, the Flame Safeguard will proceed to the Main Flame Ignition mode. After 10 seconds, the pilot burner will shut off. The UV Scanner must detect main flame at this point or the Flame Safeguard will "Lock-Out" the burner and indicate an alarm condition for the burner. This can be cleared by pressing the reset button on the Flame Safeguard. During normal operating, whenever the Operation Control reaches setpoint and opens the contacts, the main burner will safely shut down. On the next "Call for Heat", the Operating Control contacts will close and the burner will automatically start. If burner shutdown is caused by the opening of contacts in one of the safety circuits (low water, high or low gas pressure, high limit, etc.) it may be necessary to press the appropriate reset button to get the burner to restart. If it is necessary to press any reset button to operate the burner, it is recommended that the cause of the safety shutdown be determined and, if required repaired before restarting the burner.

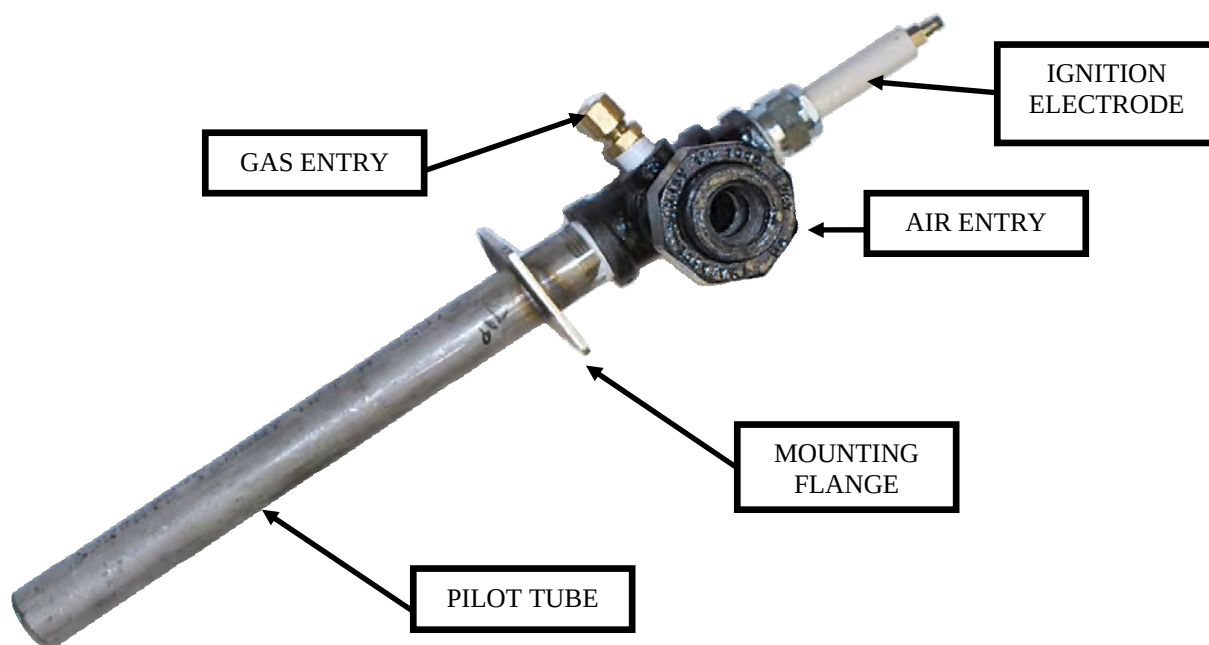
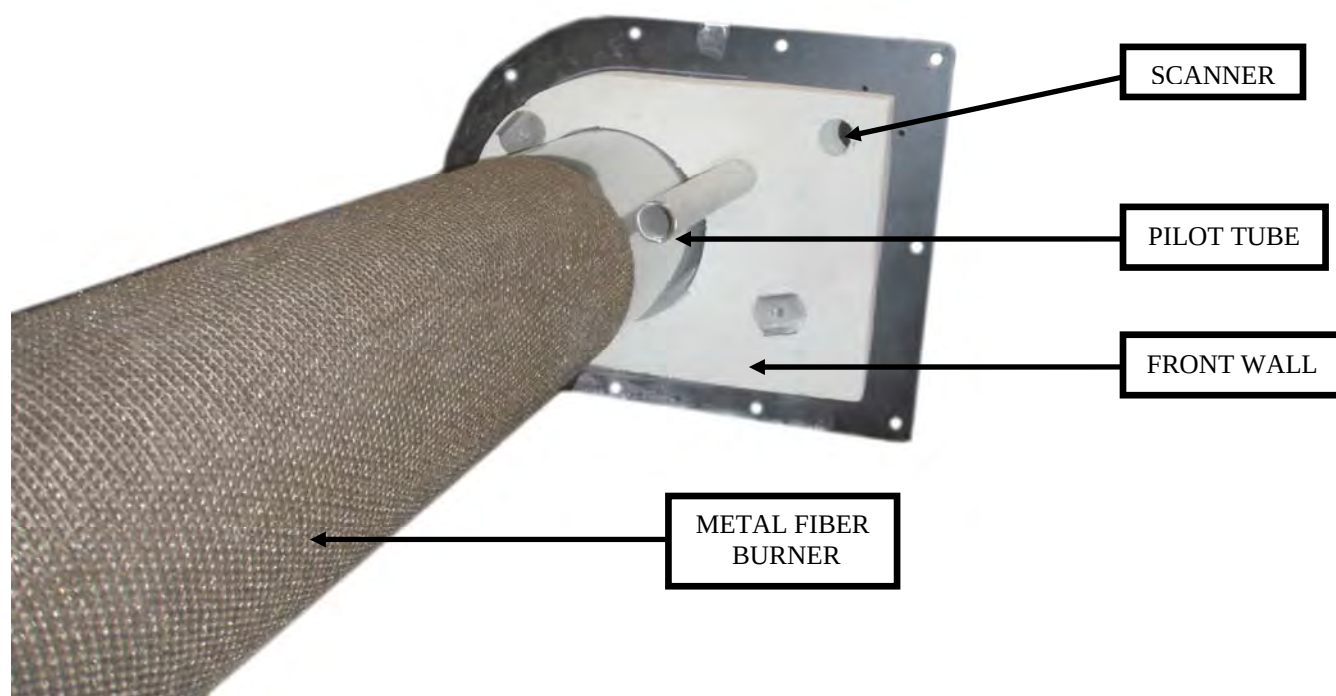
2. IGNITION SYSTEM:

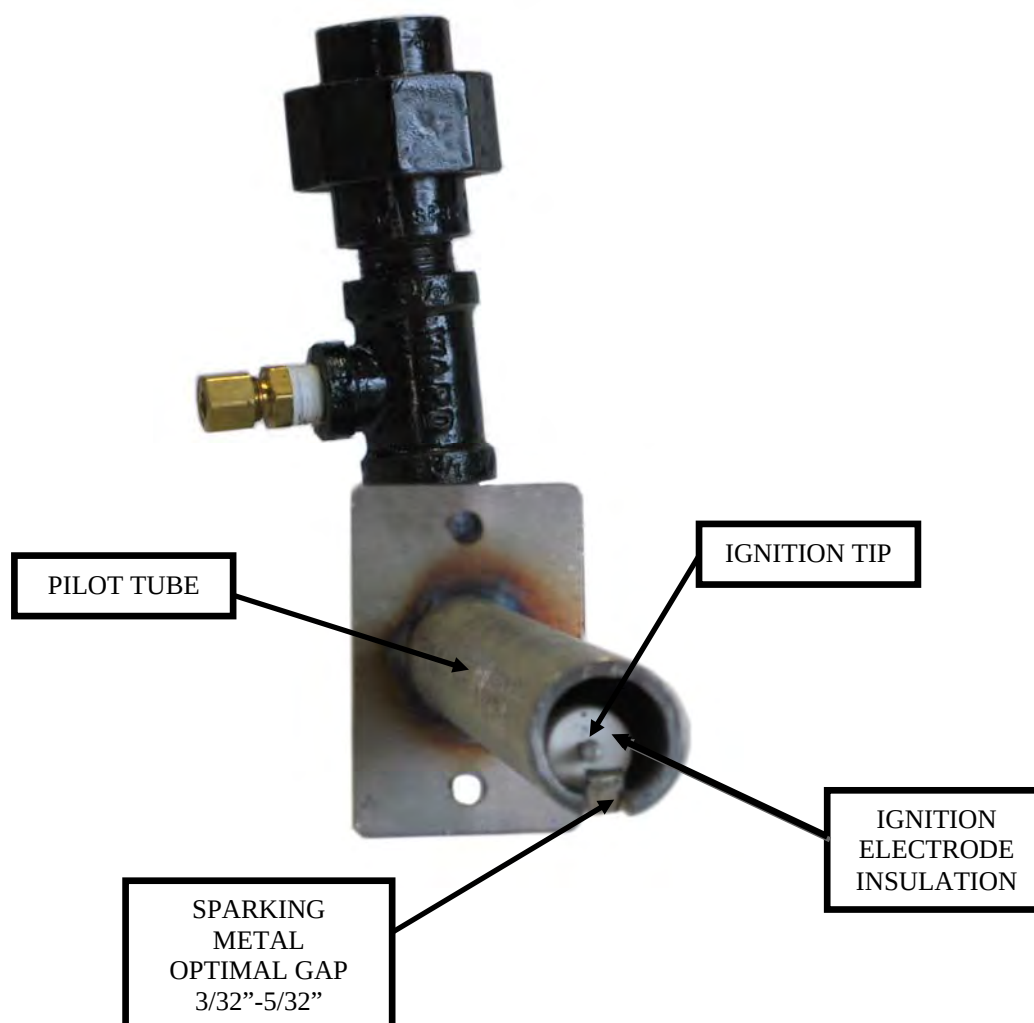
The Ignition System consists of an Electric Pilot Valve, special high voltage solid state Ignition Transformer, and a spark ignition electrode, which will automatically ignite the pilot. **NOTE: Only the factory supplied transformer or an identical one should be used.** When there is a call for heat, a pre-purge period will occur then the flame safeguard energizes the pilot valve and ignition transformer to produce an intense spark, which will readily ignite the pilot. Most systems employ a 10 second pilot trial and a 10 second main flame establishment period.



3. PILOT BURNER UV FOR CYLINDER BURNERS (WATER WALL BOILERS):

On this system, a pre mix pilot with gas and air is delivered to the pilot tube containing an igniter which sparks to the tube end. This system will energize both spark and pilot valve for 10 seconds. If a good pilot flame is sensed, main ignition occurs. There is a 10 second period for main flame ignition then the pilot and spark are terminated. The flame safeguard will supervise main flame and close fuel valves should a flame not be detected. The ignition transformer must be a Honeywell Q75A1016. With this transformer, the UV scanner will not see the ignition spark. Always test that spark only will not energize main fuel valves.





4. **ELECTRIC GAS VALVE(S):**

The Electric Gas Valve s are electrically operated valves installed on the gas line for the purpose of shutting off all gas flow to the burner except the pilot. The valves are wired for 115 volt, single phase operation. Both gas valves can only be energized if all the safety and limit controls are safe so that in the event of low water, pilot failure, main flame failure, electrical failure, or limit failure, they will automatically shut off the main flow of gas, thereby, preventing a hazardous condition from occurring in the boiler. For further details, see Bulletin on particular valve. The Siemens SKP75 series of gas valves is normally used. The valve provides safety shutoff as well as gas pressure regulation and ratio control.

5. **LOW WATER CUTOFF:**

All Boilers are furnished with a probe type low water cutoff. A separate secondary low water cutoff may be utilized. The low water cutoff will shut off the main burner on a low water experience. This control is very reliable having no moving parts on the water side. If a float type control is furnished, it should be regularly flushed and cleaned. One manual reset control is furnished. It is necessary to reset this control any time a low water condition occurs. For further details, see Bulletin on this control.

6. **OPERATING & HIGH LIMIT CONTROLS:**

All boilers are furnished with separate high limit temperature plus other limit controls and operating controls. See separate Bulletin for proper setting and adjustments of these controls.

7. **OTHER TRIM:**

The Boiler Controls Switch is furnished for convenience in electrically shutting off the boiler controls circuit. **CAUTION--HAZARD OF ELECTRIC SHOCK:** The Boiler Controls Switch does not shut off all electricity to the panel box. A Main Line Disconnect Switch, with fused disconnect or circuit breakers, must be installed and turned "OFF" whenever servicing the panel.

All Boilers Controls are standard for 115 Volt operation. When 3 Phase service (or 230 Volt, single Phase) is used, a Control Transformer may be furnished to reduce boiler control voltage to 115 Volts. The boiler control circuit is then protected from over-current and short circuits by a circuit breaker or a fuse.

If a door interlock is furnished, it turns off the boiler control circuit whenever the panel door is open. It does not shut off all electricity to the panel box. Be certain that the panel door is tightly shut so this interlock is energized or the boiler will not turn on.

WARNING: ONLY QUALIFIED SERVICE PERSONNEL SHOULD ATTEMPT TO STARTUP, ADJUST OR REPAIR THIS UNIT. NEVER BYPASS ANY SAFETY CONTROL.

3.7 – TEMPERATURE CONTROLS

PARKER BOILER CO.
HOT WATER BOILER TEMPERATURE CONTROLS

The temperature of the boiler water is controlled by adjusting the following controls. In some cases two separate temperature controllers are located in a single enclosure with two separate temperature indicators and are labeled accordingly. The temperature settings referred to below are recommended settings only. The actual settings depend upon the load and operating conditions and should be adjusted in the field for best results to match requirements of the particular application. The indicator settings of the capillary type these controls are approximations and if a more accurate control is required, these controls should be set with reference to an accurate thermometer. Temperature controls may be furnished either direct well immersion or remote bulb type.

1. MANUAL RESET HIGH LIMIT CONTROL - STANDARD ALL BOILERS:

The Manual Reset High Limit Control is standardly furnished to shut off the main burner if the temperature in the boiler rises above the setting of the operating control and it fails to operate. This control is normally located to sense the outlet water temperature and is furnished with Manual Reset. This control(s) should be set to shut off above the Operating Control at the maximum temperature setting of the system. If the Manual Reset High Limit Control shuts down the boiler, do not place the boiler back in operation until the causes of the shutdown has been determined and corrected. This control may be combined with another control, 2 bulbs in 1 well.

2. OPERATING CONTROL - STANDARD ALL BOILERS:

The Operating Control is standardly furnished to shut off the Main Burner if the temperature in the boiler rises above the desired operating temperature. This control is located to sense the outlet temperature on Hot Water Boilers.

3. MODULATING CONTROL:

The Modulating Temperature Control is wired in conjunction with the Modulating Fuel Valve or the VFD Fan Speed Control to control the temperature in the boiler by modulating the Main Burner firing. This control should be set at the desired operating temperature. If the control has an adjustable Proportioning Range, this should be set at approximately 10°F. With these settings the boiler will operate on high fire until the temperature approaches the set point, then the Main Burner will begin to modulate to the low fire setting.* If the temperature continues to rise, the Operating Control will turn off the boiler.

NOTE: Refer to detailed Bulletin or particular control for setting instructions.

